Journal of Accounting and Economics 81 (2026) 101839

Contents lists available at ScienceDirect

Journal of Accounting and Economics

journal homepage: www.journals.elsevier.com/journal-of-accounting-and-economics

Does taxpayer assistance encourage entrepreneurship?

Daphne M. Armstrong [a,*], Stephen Glaeser [b]

[a] University of Michigan, USA
[b] University of North Carolina, USA

ARTICLE INFO

JEL classification:
H23
H25
L26
M13

Keywords:
Entrepreneurship
Taxpayer assistance
Taxation
New business formation

ABSTRACT

We examine whether taxpayer assistance with tax filing and compliance affects entrepreneurship. Taxpayer Assistance Centers help taxpayers, including entrepreneurs, to correctly file their taxes and navigate the tax system. We find that Taxpayer Assistance Centers positively associate with local entrepreneur entry and with overall Schedule C business income levels. We conclude that taxpayer assistance encourages traditional business entrepreneurship by reducing compliance costs that stem from tax complications and complexity.

We appreciate helpful comments and suggestions from seminar and conference participants at the 2023 BYU Accounting Research Symposium, NHH-BU tax reading group, Ohio State University, University of North Carolina, Sungkyunkwan University, the University of Iowa, the University of Colorado Boulder, the 2024 Journal of the American Taxation Association (JATA) Conference, and the University of Toronto, as well as from an anonymous journal reviewer, Elisa Casi-Eberhard, Lisa De Simone, John Gallemore, Nathan Goldman, Michelle Hanlon, Robert Hills, Jeff Hoopes, Christian Hutzler, Hansol Jang (discussant), Ed Maydew, Will McBride, Marcel Olbert, Scott Rane, Elena Simintzi, Kevin Standridge, Bridget Stomberg, Jake Thornock, and two anonymous JATA Conference reviewers. We thank IRS staff for their timely responses to our Freedom of Information Act requests and many helpful institutional discussions.

1. Introduction

We examine whether taxpayer assistance affects entrepreneurship. Potential entrepreneurs may worry that filing business taxes will impose effort costs and create noncompliance risk (e.g., Aghion et al., 2017; Zwick, 2021). For example, surveyed entrepreneurs rank setting up and paying taxes as the third and fourth most difficult factor in starting their business, surpassing even insuring and structuring their business (Kauffman Foundation, 2018). Further, Congress regularly hears arguments that tax complexity and compliance are “special burdens” on small businesses.[1]

Given the penalties for tax noncompliance, the expense of outsourcing compliance to a third party, and the fact that outsourcing

* Corresponding author.
E-mail addresses: armsd@umich.edu (D.M. Armstrong), Stephen_glaeser@kenan-flagler.unc.edu (S. Glaeser).

[1] E.g., in 1998: https://www.govinfo.gov/content/pkg/CHRG-105hhrg52037/html/CHRG-105hhrg52037.htm, accessed November 16, 2022. In 2014: https://www.congress.gov/113/chrg/CHRG-113hhrg87461/CHRG-113hhrg87461.htm, accessed June 20, 2025. In 2025: https://edworkforce.house.gov/uploadedfiles/milito_testimony.pdf, accessed June 20, 2025.

https://doi.org/10.1016/j.jacceco.2025.101839
Received 26 April 2024; Received in revised form 9 October 2025; Accepted 14 October 2025
Available online 15 October 2025

compliance does not eliminate all tax risk, potential entrepreneurs may decide it is simpler and safer to stay at their current jobs. By helping or promising to help potential entrepreneurs with tax filing, taxpayer assistance can encourage business formation by lowering potential tax compliance costs. However, some argue that taxpayer assistance will not affect entrepreneurship because taxes are not a salient issue for most potential entrepreneurs and because the tax complexity facing entrepreneurs is overstated (e.g., Matthews, 2017; Dunkelberg, 2022). In total, whether taxpayer assistance affects entrepreneurship is an open empirical question.

To provide empirical evidence on taxpayer assistance and entrepreneurship, we examine the impact of Internal Revenue Service (IRS) Taxpayer Assistance Centers (TACs) on local entrepreneurship. The IRS does not operate TACs as part of its enforcement function. Instead, the IRS operates TACs to provide taxpayers, including entrepreneurs, in-person assistance navigating the tax system and filing their federal taxes. TACs assist with basic procedural issues that stem from a lack of taxpayer awareness, such as choosing which forms to use, but also with issues that surveyed traditional business entrepreneurs find complex, such as payroll taxes and tax law (e.g., Beers et al., 2012; Kauffman Foundation, 2018).

Our focus on taxpayer assistance mirrors the IRS's 2023 focus on providing "[a] world-class customer service operation" that assists taxpayers with "navigating complex tax laws and accessing the credits they deserve."[2] Given that TACs provide assistance with tax forms and answer questions about tax law, we expect them to alleviate the burden of tax compliance and filing. Tax assistance is in high demand, and much of this demand goes unmet by other federal services.[3] We obtain operating data for over 400 TACs using a Freedom of Information Act (FOIA) request to the IRS. We examine the impact of TAC operations on entrepreneurship within the same ZIP code. We expect TACs to impact entrepreneurship particularly within the same ZIP code because (1) TACs provide mostly in-person services, (2) local potential entrepreneurs are most likely to be aware of local offices, and (3) the IRS does not broadly advertise TACs (US Department of the Treasury, 2023).[4]

To control for local business conditions, demographics, and state-level factors such as state tax rates, we separately include metropolitan statistical area (MSA)-year and county-year fixed effects, along with controls for demographic characteristics the IRS uses to select TAC locations. We measure entrepreneurship using data on new business registrations collected by the Startup Cartography Project.[5] These new businesses are almost all traditional business ventures that will likely remain small businesses, such as cafes and auto garages. Consequently, our results likely do not stem from innovation-driven entrepreneurship, especially because innovation-driven businesses are less sensitive to taxes, typically have the resources to navigate tax compliance, and focus largely on developing risky ideas rather than on short-term cash flows.[6] While any individual small business is unlikely to be as important as large, innovation-driven businesses, traditional small businesses account for approximately 45 % of US GDP and 46 % of private employment (Kobe and Schwinn, 2018; US Chamber of Commerce, 2023).[7] Despite the economic importance of traditional small businesses in the aggregate, they may be particularly sensitive to taxpayer assistance due to their modest individual sizes.

Consistent with taxpayer assistance significantly encouraging business formation, differences-in-differences estimates suggest that TACs associate with significant increases in local entrepreneur entry. The 90 % confidence interval of the estimate in our most stringent specification suggests that TACs associate with a 4 %–13 % increase in entrepreneurship; 9 % at the midpoint. Starting from the median of entrepreneur entry of 114, a 9 % increase corresponds to the creation of about 10 businesses. This TAC-related increase is about a third of the effect generated by access to health insurance (Holtz-Eakin et al., 1996) and about double the effect of access to gig economies (Barrios et al., 2022) or of the right to *borrow* $30,000 against the entrepreneur's home (Jensen et al., 2014, 2022). We believe our estimate is reasonable in light of these prior estimates, given that the cost of outsourcing tax work is several hundred to several thousand dollars each year at a minimum.[8]

One potential concern with using TACs as a source of variation is the possibility that the IRS opens or closes TACs based on omitted characteristics that vary across ZIP codes and within county- and MSA-years in a way that biases the results. For example, the IRS may choose to open TACs in locations with expected future increases in business activity. However, IRS materials suggest that the agency does not consider future trends in entrepreneurship, but rather historical location demographics, when selecting TAC locations.[9]

[2] Available at https://www.irs.gov/newsroom/written-testimony-of-daniel-werfel-commissioner-internal-revenue-service-before-the-hoac-subcommittees-on-government-operations-and-the-federal-workforce-and-health-care-and-financial-services-and-irs-operations, accessed November 14, 2023. These sources of assistance include operating more TAC locations.

[3] For the 2022 filing season, the IRS received more than 72.8 million phone calls for help, but representatives answered only 10 %; https://www.taxpayeradvocate.irs.gov/reports/2023-objectives-report-to-congress/full-report/, accessed November 17, 2022. The IRS even lists tax topics that it will not assist with over the phone and notes that it "can't offer line-by-line help with any form; " https://www.irs.gov/help/find-information-on-complex-tax-topics, accessed November 17, 2022. During the sample period, taxpayers did not have to schedule TAC appointments by phone (e.g., taxpayers could walk into TACs). This protocol changed in IRS fiscal year 2017; https://www.taxpayeradvocate.irs.gov/wp-content/uploads/2020/10/ARC17_Volume1_MSP_010_TAC.pdf, accessed June 21, 2023.

[4] ZIP codes cover a relatively small geographic area. In subsequent analyses we examine spillovers from TACs to nearby ZIP codes and find that TAC effects rapidly dissipate. This rapid dissipation suggests either that TACs are salient mainly to potential entrepreneurs who live or work in the same ZIP code as the TAC and/or that potential entrepreneurs' travel costs are quite high. We speculate that the former explanation is more likely.

[5] E.g., Barrios et al. (2021), Botelho et al. (2021), Andrews et al. (2022), and Barrios et al. (2022).

[6] E.g., Bankman (1994), Goldberg (2001), Johnson (2009), Hall and Woodward (2010) and Allen et al. (2023).

[7] In 2023, even very small firms with one to four employees accounted for 4.8 % of US private employment; https://www.bls.gov/web/cewbd/table_f.txt, accessed April 23, 2024. In untabulated tests we find no relation between taxpayer assistance and innovation-driven entrepreneurship.

[8] Since most traditional businesses produce modest profits, several thousand or even several hundred dollars can matter along a significant margin; recall that the entrepreneur considers the *marginal* utility of starting their own business.

[9] Available at https://www.irs.gov/pub/irs-news/tac-criteria.pdf, accessed June 21, 2023.

Further, the IRS also takes several years to open or close a TAC. For example, in 2005 the IRS used 2004 operating and demographic data to select TACs to close and began closing these TACs in earnest in 2011 (see Section 2.1 for details). Consequently, any endogenous variable considered by the IRS should result in differential trends in entrepreneurship several years prior to the opening or closing of a TAC.

Contrary to endogeneity driving the relation between TACs and entrepreneurship, we find no evidence that entrepreneurship varies in the years prior to changes in TAC activity (i.e., we find no evidence of pre-existing differential trends in entrepreneurship within a ZIP code prior to changes in TAC activity). Further, IRS materials specify that the agency selects TAC locations based in part on many unobservable characteristics, such as furniture costs, which alleviate broader endogeneity concerns as these unobservable characteristics are plausibly exogenous with respect to entrepreneurship.

We estimate several extensions of the main results. We first examine whether TACs additionally affect entrepreneurship in nearby ZIP codes. Consistent with spillovers from TACs to nearby ZIP codes, we find that TACs relate to entrepreneurship in nearby ZIP codes by approximately one-fourth as much as they relate to entrepreneurship in the same ZIP code. The relatively small magnitude of these spillovers supports examining the impact of TACs at the ZIP code level. Further consistent with salience effects, we find that TACs that remained open throughout the sample period significantly related to entrepreneurship in nearby ZIP codes.

We also examine several alternative measures of business activity in light of the potential concern that TACs increase business registrations without a commensurate increase in real business activity. For example, a potential concern is that TAC workers advise existing entrepreneurs who report their business income on their personal tax returns to register their business, increasing business registrations but not underlying entrepreneurial activity. We view this concern as unlikely because TAC workers are not supposed to provide non-tax advice. Nonetheless, we examine the ways in which TACs relate to business income reported to the IRS (Schedule C income). If TACs only cause individuals to register their business for which they previously reported income, then we expect TACs to not affect Schedule C income. In contrast, we expect TACs to positively relate to Schedule C income if TACs increase business activity. Consistent with TACs increasing real business activity, we find that TACs associate with a 7 % increase in Schedule C income.

Another potential concern is that TACs drive business registration by illegal businesses that did not report income (i.e., transitions from the informal to the formal economy without changes in real activity). We also view this concern as unlikely because the IRS operates TACs independently of its enforcement function.[10] Nonetheless, in light of this concern we examine the ways in which TACs relate to Census establishments, which are physical locations where business is conducted. Because the Census measures establishments from state unemployment and state census data, and not via IRS data, Census establishments are less likely to reflect underreporting to the IRS. We find that TACs relate to a 1.3 % increase the number of establishments in a county, inconsistent with the concern that TACs drive business registration by previously illegal businesses.[11]

We next examine how startup activity evolves after TAC openings and closings. We expect TAC openings to have a delayed effect on entrepreneurship as potential entrepreneurs become aware of the TAC over time and prepare potential businesses. Consistent with this expectation, we find that TAC openings increase entrepreneurship at a horizon of three years. In contrast, TAC closings immediately decrease entrepreneurship, consistent with potential entrepreneurs forgoing business creation when losing assistance.

We further extend the main results by examining whether the state tax environment moderates the relation between TACs and entrepreneurship. Although TACs do not directly assist with state taxes, TACs should assist with most/all tax compliance in states that do not tax business income or that enforce tax rules that closely follow the federal rules (e.g., North Carolina or Texas). In contrast, TACs will leave a great number of compliance issues unaddressed in states where the state tax rules differ substantially from the federal rules (e.g., California). Therefore, the level of conformity between state and federal taxes should moderate the effect of TACs on entrepreneurship. Consistent with this reasoning, we find that TACs relate to entrepreneurship mainly in states without state taxes or that begin state tax calculations with federal adjusted gross income.

Finally, we examine the robustness of the results to estimating a “stacked” differences-in-differences regression in light of the potential concern that the staggering of TAC treatments biases estimates due to heterogeneous treatment effects.[12] We find similar results, likely because of the large number of never-treated control observations in the sample and the relatively short panel (De Chaisemartin and d’Haultfœuille, 2020).

We contribute to the academic literature and policy debates by documenting the ways in which taxpayer assistance relates to traditional business entrepreneurship, an important real outcome (Lester and Olbert, 2025).[13] Despite the significance of traditional businesses in the aggregate, traditional business entrepreneurship has declined (Meyer, 2011; DePillis, 2022). The evidence that taxpayer assistance encourages traditional business entrepreneurship may inform policymakers interested in reversing this decline. However, recent funding changes are transforming the service operations of the IRS, including taxpayer assistance options (Holtzblatt, 2025). The results suggest that reductions in taxpayer assistance would reduce traditional business entrepreneurship. However, the

[10] In 2023 only 4 % of TACs operated in the same ZIP code as an IRS division office (which handles enforcement of larger businesses). The IRS was unwilling to provide us with historical data on IRS enforcement offices due to privacy concerns.

[11] We attribute the smaller effect size of TACs on Census establishments to examining effects at the broader county level, which is necessary because the Census does not track establishments at the ZIP code level.

[12] E.g., Gormley and Matsa (2011), De Chaisemartin and d’Haultfœuille (2020), Barrios (2021), and Baker et al. (2022).

[13] Related prior work includes Amberger et al. (2023), which examines how country level tax complexity, as measured by the amount of time that surveyed medium-sized firms indicate they spend preparing, filing, and paying taxes, relates to firm-level investment. Denes et al. (2023) find that angel investor tax credits do not relate to entrepreneurial activity as measured by startup outcomes (e.g., venture capital funding). Aobdia et al. (2024) find that access to high-quality accounting labor improves startup outcomes.

results also suggest that simplifying tax filing and compliance by reducing tax complications and complexity would likely encourage entrepreneurship. Moreover, the evidence that TACs mainly affect local entrepreneurship due to limited awareness suggests that improving awareness of TACs in neighboring geographic areas might be a low-budget alternative to expanding operations.

We organize the rest of the paper as follows. Section 2 provides background and institutional details. Section 3 discusses the empirical approach and results. Section 4 concludes.

2. Institutional detail and prior literature

2.1. Taxpayer Assistance Centers

TACs assist taxpayers, including business owners, with tax forms, questions about tax law, navigating issues with the IRS, recommending changes to avoid issues with the IRS, receiving all eligible deductions and refunds, etc.[14] Fig. 1 presents data on the most common topics with which taxpayers request assistance, based on the help codes for a random sample of TACs provided by the IRS. TACs assist with business-specific items such as questions about tax law and tax returns in 4.2 % of visits, individual-specific items in 0.4 %, and ambiguous items (where it is not clear whether assistance is given with business or with individual items) in the remaining 95.4 %. The most common ambiguous assistance items are tax related transcripts (e.g., requests for specific line items from prior returns), balance due and collection notices, non-cash payments (e.g., payments via checks), and non-tax related transcripts (e.g., requests for IRS information returns such as the W-2).[15] Based on our anecdotal conversations with TAC workers, most visitors seek help with personal tax issues, although help with business issues is not unusual.

As noted by Bradford (1986), "The hardest [tax] problems have to do with financial affairs and directly owned businesses …. [O]ne of the irritations in life is determining which outlays are eligible for deduction … consider a classic problem of tax administration: the office at home. Are all or any of the expenses deductible? … Home computers and automobiles used partly for business purposes are further examples … The complexity involved in the tax treatment of such transactions is largely the result of the narrow line dividing expenses for purposes of earning a return and those for direct personal benefit. The necessary distinctions create headaches for taxpayers" (p. 270–271). TACs can assist individuals navigating these and other potential tax compliance issues and thereby encourage marginal potential entrepreneurs to start a business by providing assistance *ex ante* or the knowledge that they will receive assistance *ex post*. However, we stress that TACs do not assist with issues that legal academics, tax academics, or the tax directors of public firms would find complex, such as transfer pricing.

The IRS operates TACs independently of its enforcement function. TAC location decisions are generally a slow-moving, bureaucratic process. For example, in 2005 the Government Accountability Office (GAO) assessed the fiscal year (FY) 2006 IRS budget request and advised that, "In light of the current budget environment and IRS's improvements in taxpayer service over the last several years, this is an opportune time to reconsider the menu of services it provides. It may be possible to maintain the overall level of assistance to taxpayers by changing the menu of services offered, offsetting reductions in some areas with new and improved service in other areas."[16] In response to the GAO's assessment, then IRS commissioner Mark W. Everson released a statement on the expected closure of 68 TACs, a process that began in earnest in 2011 and even then took several years.[17] In selecting which TACs to close, Everson noted that the IRS's methodology relied on FY2004 TAC operating and Census data.

The IRS's methodology for closing TACs highlights four important institutional details for the research design. First, the methodology suggests that the IRS does not randomly operate TACs and that we must exercise caution with respect to potential selection effects and correlated omitted variables. Second, the IRS's methodology suggests that the IRS selects TAC locations based on historical characteristics rather than concurrent changes in activity. Consistent with this, anecdotal evidence suggests that the IRS does not target areas with booming economic activity for TAC operation.[18] Because the IRS does not select TAC locations based on concurrent changes in activity, selection is unlikely to drive any correlation between changes in TACs and contemporaneous entrepreneurship. Third, the slow-moving nature of TAC location decisions suggests that any endogeneity between TAC location decisions and entrepreneurship is likely to manifest prior to changes in TAC locations. Fourth, many of the TAC operation characteristics considered by the IRS are largely orthogonal to entrepreneurship outside of their effect on TAC operations (e.g., furniture costs), suggesting that there is a large degree of exogeneity in TAC locations.[19]

During the sample period, individuals could schedule TAC assistance in advance or receive walk-in assistance, including with tax law. Unfortunately, the IRS was unable to provide systematic information on historical TAC wait times, but modern audits suggest that

[14] Available at https://www.irs.gov/pub/irs-pdf/p334.pdf, accessed June 26, 2023.

[15] Fig. 1 reports every item that is ambiguous if it is above 5 % of the total and aggregates the remaining items into "Other ambiguous" for parsimony.

[16] Available at https://www.gao.gov/assets/gao-05-566.pdf; https://www.taxpayeradvocate.irs.gov/wp-content/uploads/2020/08/ARC17_ExecSummary.pdf, accessed June 25, 2023.

[17] Available at https://www.irs.gov/pub/irs-news/tac-statement.pdf, accessed June 21, 2023.Available at https://www.nbcnews.com/id/wbna8007103; https://www.nytimes.com/2005/05/28/business/irs-closing-help-centers-in-some-areas.html.

[18] Unfortunately, we cannot observe these orthogonal characteristics and the IRS was unable to provide data on them.

[19] Available at https://www.tigta.gov/sites/default/files/reports/2024-05/2024100022fr.pdf; https://home.treasury.gov/news/featured-stories/continuing-improvements-to-irs-customer-service-in-filing-season-2024, accessed February 8, 2025. Modern audits suggest in-person wait times were usually between 0 and 105 min.

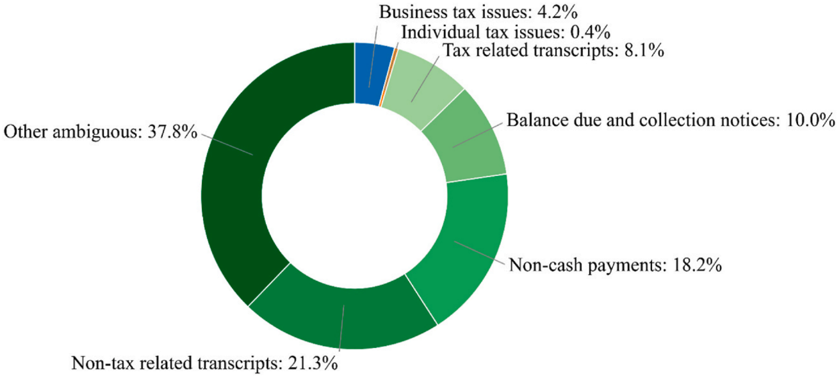

Fig. 1. TAC assistance by topic
This graph presents the fraction of TAC visits corresponding to individual tax issues (blue), business tax issues (orange), or various ambiguous tax topics (shades of green), from data on TAC help codes collected by the IRS. The most frequent ambiguous assistance items are tax related transcripts (e.g., taxpayer requests for certain line items from prior tax returns and general account inquiries), balance due and collection notices, non-cash payments (e.g., payments via checks and money orders), and non-tax related transcripts (e.g., taxpayer requests for IRS forms received via information returns such as the W-2, 1098, and 1099; and personal account inquiries). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the wait time to arrange a TAC appointment approximates that to receive toll-free phone help from the IRS.[20] However, consistent with TACs enjoying a comparative advantage to alternative help options from the IRS, they are well trafficked, further suggesting that TACs materially affect constituents. Over the three-year period 2014–2016, the average TAC experienced 13,395 visits annually. The totals include 5,449,445 visits to 389 TACs in 2014, 5,434,144 visits to 378 TACs in 2015, and 4,426,918 visits to 376 TACs in 2016.[21]

In addition to TACs, individuals seeking free IRS tax advice can use toll-free help lines or Volunteer Income Tax Assistance (VITA) programs. Toll-free help lines mainly assist taxpayers with understanding tax law and meeting tax obligations. In contrast, TACs target taxpayers who prefer in-person assistance and who need help to resolve tax issues, apply tax law to individual taxpayer situations, make payments, etc. VITA assists taxpayers in preparing their tax returns, but taxpayers must be low-income individuals. Additionally, VITA offers limited tax assistance with self-employment income and will not prepare certain types of Schedule C returns (i.e., those with losses, depreciation, or business use of home). VITAs are also staffed by volunteers and most are unavailable after April 15. In total, TACs offer more assistance options, and with business income issues in particular. Surveys suggest that individual taxpayers prefer self-assistance options (e.g., the IRS website) for transactional tasks, but prefer assisted services (e.g., TACs) for "more complex interactive tasks like responding to a notice" (National Taxpayer Advocate, 2012, p.253). Further, the National Taxpayer Advocate (2016) notes that TAC users are "especially loyal to the TAC" relative to users of other free IRS services (p. 17).

Several anecdotes support the assumption that TACs offer superior, or at least different, help than other free service options. In an interview, Carl Aley, Director of Management and Exempt Organization Audits for the Treasury Inspector General for Tax Administration, stated, "[T]he reason that the IRS [now schedules in-person TAC visits over the phone] is, in a lot of cases the IRS is able to resolve the taxpayer's issue over the phone without the need for them to actually go in and visit a TAC. So that's why they do it that way. In fact, in 2023 … a little over 30 % of all callers to the TAC … were able to resolve their issue without the need to schedule an appointment."[22] This suggests that in about 70 % of potential cases TACs helped with issues that toll-free help lines could not. Similarly, Nina Olson, National Taxpayer Advocate at the IRS, testified to Congress about IRS plans to close some TACs and stated that the IRS "overestimates taxpayers' ability or willingness to conduct complex financial transactions in an electronic or self-service format."[23]

Although TACs are not geographically constrained in whom they assist, we expect them to particularly impact geographically proximate individuals who are most likely to be aware of, and who have the shortest travel time to, the TAC. TAC openings and closings are not well-advertised; the Treasury Inspector General for Tax Administration criticized the IRS for not providing sufficient information, particularly about openings and closings (US Department of the Treasury, 2023). Further, a Factiva search for "Taxpayer

[20] Because the IRS operates on a 9/30 fiscal year end, these estimates are also over these fiscal year ends (e.g., the 5.4 million visits in fiscal year 2014 occurred between October 1, 2013 and September 30, 2014). Available at https://www.taxpayeradvocate.irs.gov/wp-content/uploads/2020/10/ARC17_Volume1_MSP_010_TAC.pdf, accessed June 25, 2023.

[21] Available at: https://federalnewsnetwork.com/workforce/2024/06/irs-taxpayer-assistance-centers-still-have-some-work-to-do-on-the-customer-service-front/, accessed July 29, 2025.

[22] Available at: https://www.taxpayeradvocate.irs.gov/wp-content/uploads/2020/09/finalntawrittentestimony2005jointrra98reviewhearing.pdf, accessed July 29, 2025.

[23] E.g., https://www.nbcrightnow.com/news/taxpayer-assistance-center-now-open-at-new-location-in-union-gap/article_3d29475c-c6d4-11ef-8562-fbdc4ab8f918.html. If we expand the search to include non-"All Web News," we identify over 2000 documents, but all are Congressional documents, law reviews, the Federal Register, and other articles not typically accessed by the general public.

Assistance Centers" and similar words identified only ten TAC-focused articles.[24] Consequently, the effect of TACs is likely driven primarily by individuals seeing the TAC or hearing about it via word of mouth. To the extent that TACs affect non-proximate individuals, and therefore outcomes at control observations, the resulting estimates will underestimate the benefits of TACs. We model these potential spillovers and find evidence that they lead to marginally conservative estimates.

2.2. Prior literature on tax compliance, complications, and complexity

Taxpayer assistance with compliance and filing may encourage entrepreneurship by resolving aspects of the tax system that are complicated and complex from the perspective of traditional business entrepreneurs. Kamensky (2011) defines complicated problems as those where "one can usually predict outcomes by knowing the starting conditions," and complex problems as those where "the same starting conditions can produce different outcomes, depending on interactions of the elements in the system." For example, knowing which tax forms to use is complicated while figuring out how to optimally respond to an IRS notice can be complex.

Prior studies find that complicated and complex tax problems can impose significant burdens on businesses by generating compliance and filing costs. Amberger et al. (2023) find that country-level tax complexity negatively relates to firm-level investment. Aghion et al. (2017) find that self-employed individuals "leave money on the table" due to tax complexity and value tax simplicity by up to 650 euros a year. Benzarti (2020) finds that individuals forgo large tax savings to avoid tax complications and complexity. Zwick (2021) finds that most eligible firms do not claim refunds for tax losses, suggesting that tax complications and complexity cause businesses to forgo valuable deductions. Feldman et al. (2016) find that individuals reduce their reported wage income when losing a child tax credit, suggesting that individuals find calculating the child tax credit complicated. Cowx (2021) finds that firms respond less to R&D tax credits when enforcement uncertainty is greater, suggesting that tax uncertainty can hinder innovation incentives. Weber (2015) finds that a one standard deviation increase in state tax code length associates with less business dynamism.

Collectively, prior work suggests that complicated and complex tax problems can impose significant tax compliance and filing costs on businesses and individuals. While our tests cannot delineate between tax complication and complexity, they shed light on how taxpayer assistance with these issues affects entrepreneurship.

3. Empirical approach and results

3.1. Regression model, sample, and descriptive statistics

We examine how local entrepreneurship varies as a function of TAC presence using the following staggered differences-in-differences model:

$$ihs(Startups_{i,t}) = \beta_0 + \beta_1 TAC_{i,t} + \gamma' X_{i,t} + Y_i + Z_{j,t} + \varepsilon_{i,t,} \tag{1}$$

where i indexes ZIP codes, t indexes calendar years, and j indexes counties or MSAs. *Startups* is the number of for-profit business registrations recorded by the Startup Cartography Project in ZIP code i at time t (Barrios et al., 2021, 2022; Andrews et al., 2022); *ihs* refers to the inverse hyperbolic sine transformation.

The Startup Cartography Project measures startups using state business registrations. All corporations, partnerships, and limited liability companies register with the Secretary of State or equivalent to enjoy the legal benefits of business creation. While it is possible to operate as a sole proprietorship and forgo registration, doing so forgoes the legal benefits. Given that single-member LLCs have the same default treatment as sole proprietors and TAC personnel are not supposed to provide non-tax advice, exposure to TACs should not affect business registrations outside of their effect on real business activity (we further validate this assumption in subsequent tests).

We take the *ihs* of *Startups* due to skew and because we expect TACs to affect new business creation proportionally.[25] The *ihs* transformation leads to similar interpretations, and similarly reduces the influences of outliers and skew, as the natural logarithm transformation (e.g., Johnson, 1949; Burbidge et al., 1988; Glaeser and Omartian, 2022). The relative benefit of the *ihs* transformation is that it is defined for nonpositive values (e.g., zero). Although the Startup Cartography Project does not record zeros, we interpolate them for observations bookended by positive values (e.g., if for a given ZIP code the Startup Cartography Project records one new business registration in 2013, nothing for 2014, and two new business registrations in 2015, we assume the ZIP code experienced zero new business registrations in 2014).

Startups is the number of business registrations, which are primarily traditional, small businesses, not the high-growth innovation-driven businesses that garner most media attention.[26] This is consistent with our theoretical constructs, as TACs likely encourage the marginal entrepreneur who lacks the resources to obtain comparable taxpayer assistance elsewhere (the marginal entrepreneur is most

[24] For example, we expect TACs to increase entrepreneurship proportionally by 10 % to the baseline rate (e.g., 100 startups to 110 and 1000 startups to 1100) as opposed to linearly by 10 (e.g., 100 startups to 110 and 1000 startups to 1010).

[25] In untabulated results, we find that the average zip code in our sample contains 0.07 innovation-driven businesses, compared to 168 traditional businesses.

[26] Bankman (1994), Goldberg (2001), Johnson (2009), and Allen et al. (2023). In untabulated analyses, we find no significant relation between innovation-driven entrepreneurship at the zip code level, as measured by Guzman and Stern (2015), and the presence of a TAC when estimating Eq. (1); t-statistics of 0.35 and 0.41, although we note that innovation-driven entrepreneurship concentrates heavily in a few locales, many of which lack TACs.

likely self-employed or operating a traditional small business). In contrast, innovation-driven entrepreneurs should have the resources to obtain taxpayer assistance even absent a TAC.[27] One might argue that researchers should only be interested in innovation-driven entrepreneurs because they generate more expected growth than traditional entrepreneurs (Botelho et al., 2021). We do not agree. In the aggregate, traditional entrepreneurs likely generate economic activity comparable to that of innovation-driven entrepreneurs, since there are significantly more traditional entrepreneurs (e.g., for every Google and its many failed competitors there are thousands of retail shops, tradespeople, restaurants, etc.; Kobe and Schwinn, 2018; US Chamber of Commerce, 2023).[28]

TAC is an indicator equal to one if the IRS operates a TAC in ZIP code *i* at time *t*. ZIP codes cover relatively small geographic areas, suggesting that selecting a larger geographic footprint to measure the effects of TACs might increase power. However, in subsequent tests we find that the effect of TACs rapidly dissipates with distance, suggesting that TACs primarily affect local ZIP codes where travel costs are lower and salience is greater.

We use a FOIA request to the IRS to obtain TAC operating data from 2010 through 2023; our sample period extends only to the end of 2015, due to changes in TAC policies. We end the sample in 2015 because in 2016 the IRS changed several TAC policies, such as reducing staffing, hours, and services, including no longer assisting with tax law.[29] In response to our request, the IRS provided a list of every operational TAC as of January 1 of each year but could not provide exact opening and closing dates. Consequently, we cannot determine whether a closed TAC operated for most of the last year in which it is included on the IRS's list. For example, we cannot determine whether a TAC on the 2013 list but not the 2014 list was last operational as of the beginning or the end of 2013, or sometime in between. Anecdotally, our understanding is that the IRS tries to operate closing TACs through April 15 due to tax season but closes them shortly thereafter. Since April 15 is only 3.5 months into the year, we set *TAC* equal to 0 for TACs in their last year on the list.

Fig. 2 provides a map of TAC locations during the sample period. Black dots denote TACs that operated continuously during the sample period. Red boxes denote TACs that closed during the sample period. Green pluses denote TACs that opened during the sample period. Blue diamonds denote the 12 TACs that both opened and closed during the sample period. Including these 12, 35 TACs opened and 69 TACs closed during the sample period. Fig. 2 highlights that the IRS operates TACs in high-population areas, but also broadly geographically due to quasi-exogenous political requirements (e.g., a requirement that two TACs operate in every state). Fig. 2 also highlights that some TAC openings and closings occur in proximate ZIP codes.[30]

Eq. (1) includes the vector *X* of controls, which we select based on the demographic characteristics that the IRS considers when making TAC operation decisions, to the extent possible (see Appendix A for variable definitions).[31] To control for population size and wealth in a ZIP code year, we include the number of returns that claim the earned income tax credit (*Number of EITCs*), total number of exemptions filed (*Number of exemptions*), total number of tax returns (*Number of returns*), number of tax returns that report unemployment compensation (*Number unemployed*), total population (*Population*), and total reported taxable income (*Taxable income*). To further control for population effects, we follow Barrios et al. (2022) and require a minimum population of 10,000 for inclusion in the final sample. To control for racial representation, we include the populations of the Equal Employment Opportunity Commission racial groups (e.g., *Population Asian*; the omitted category is two or more races). Finally, we control for the population 65 years of age or older (*Population 65+*). We obtain all demographic controls from Census.gov and individual income tax controls from the IRS Statistics of Income.

To control for time invariant differences across ZIP codes, Eq. (1) includes a vector of ZIP code effects, *Y*. These ZIP code fixed effects isolate time-series variation in business formation and TAC operation. Isolating time-series variation is important because the discussion in Section 2.1 highlights that endogenous selection is unlikely to bias the time-series relation between TACs and entrepreneurship. Eq. (1) also separately includes vectors of county-year and MSA-year fixed effects, *Z*, to control for time-varying local

[27] Consistent with this view, policymakers are often explicitly interested in traditional entrepreneurship; in addition, prior work examines settings and forces that likely do not affect innovation-driven entrepreneurship.

[28] Some of these changes, such as requiring appointments, could have screened out taxpayers with basic issues, in turn making TACs more efficient. In untabulated results, we re-estimate Eq. (1) after extending the sample and including the interaction between TAC and an indicator for 2016, *TAC* × *2016*. The main effect on TAC is similar in magnitude and significance as its counterparts in Table 3. The coefficient on *TAC* × *2016* when including MSA-year fixed effects is small and statistically insignificant, inconsistent with a differential effect of TACs in 2016. However, the coefficient on *TAC* × *2016* county-year fixed effects is negative and statistically significant. The magnitude of the coefficient on *TAC* × *2016* relative to the coefficient on TAC suggests that TACs spurred about 35 % less entrepreneurship in 2016 than they did in prior years. This decline follows a general decline in taxpayer visits to TACs around the same time, from 5.4 million in FY2014 and FY2015 to 4.4 million in FY2016 and 3.2 million in FY2017 (Taxpayer Advocate Service, 2018).

[29] These proximate events may attenuate estimates to the extent that there are spillovers between proximate ZIP codes (e.g., because the negative effect of a TAC closing in one ZIP code is sometimes partially offset by the positive spillover of a TAC opening in an adjacent ZIP code). To explore the possibility that proximate treatment events attenuate prior estimates on TAC, we re-estimate Eq. (1) after excluding the twenty ZIP codes (ten pairs) that experience proximate treatment events (defined as those events occurring in two ZIP codes within ten miles of each other). In untabulated results, we find that the coefficients on *TAC* are statistically significant and slightly larger than their counterparts for the main results in Table 3 (estimates of 0.203 and 0.114 compared to 0.159 and 0.086). In total, we conclude that the inclusion of proximate treatment events does not drive inferences.

[30] Available at https://www.irs.gov/pub/irs-news/tac-criteria.pdf, accessed June 21, 2023.

[31] If a ZIP code spans multiple counties, we assign that ZIP code to the county that contains the highest percentage of the ZIP code's population.

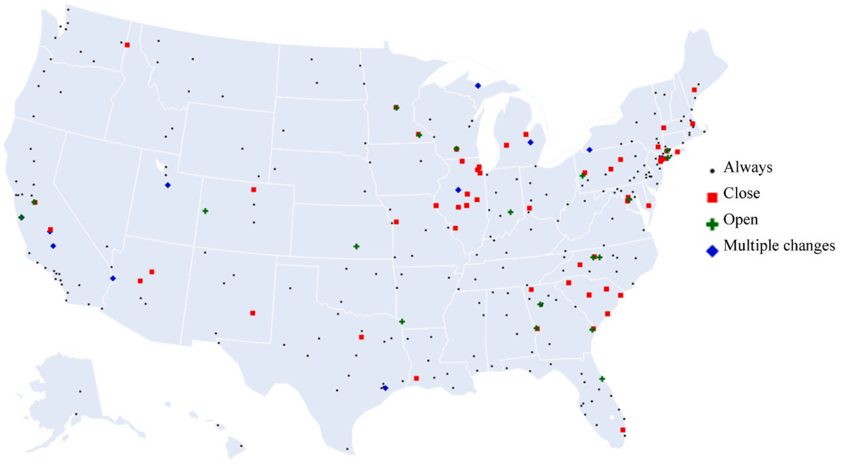

Fig. 2. TAC location map
This figure maps TAC locations in the sample. Black dots denote TACs that operate continuously over the sample period, red boxes denote TACs that closed during the sample period, green pluses denote TACs that opened during the sample period, and blue diamonds denote TACs that both closed and opened during the sample period. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

conditions and factors that may also affect entrepreneurship (e.g., local economic booms and state taxes; see Gentry and Hubbard, 2000).[32] County-year fixed effects control for variation more proximate to the TAC ZIP code on average, which could increase the precision of estimates but could also attenuate estimates due to spillovers from TACs to nearby ZIP codes within the county. Any such attenuation should be less pronounced when using the broader MSA-year fixed effects. We cluster standard errors by ZIP code due to potential serial dependence within ZIP codes over time. The final sample spans 2010 through 2015.

Table 1 reports descriptive statistics for the sample. Panel A reports statistics for untransformed variables and Panel B reports statistics for variables after the *ihs* transformation. TACs are uncommon; they operate in 3.2 % of sample ZIP code-year observations. On average, 168 new startups register in a given ZIP code each year, although there is significant variation in startup activity (standard deviation of 172). The large number of average startups in each ZIP code suggests that the *ihs* transformation is appropriate (Bellemare and Wichman, 2020).

3.2. Determinants of TAC operation

We begin by examining the determinants of TAC operation. To do so, we estimate Eq. (1) after removing *TAC* as an independent variable and replacing *Startups* with *TAC* as the dependent variable. We report results in Table 2. Column (1) reports results when including only year fixed effects. Consequently, cross-sectional differences in demographics across ZIP codes are the main source of identifying variation in column (1). In the cross-section, most ZIP code demographics statistically significantly relate to *TAC*, likely because the IRS explicitly considers these demographics when making TAC operation decisions. However, the adjusted R-squared in column (1) is only about 2 %, confirming the discussion in Section 2.1 that the IRS considers many idiosyncratic factors when making TAC operation decisions.

We next explore potential sources of bias by estimating the time-series relation between TAC operation and ZIP code demographics. To do so, we include ZIP code fixed effects and report results in column (2) of Table 2. In column (2), only one of the 13 demographic characteristics statistically significantly relates to TAC operation, which is about what one would expect by chance. We continue to find little relation between TAC operation and ZIP code demographics when including MSA-year and county-year fixed effects in columns (3) and (4). The lack of a significant relation between time-series variation in demographic characteristics and TAC operation confirms the discussion in Section 2.1 that the IRS does not consider contemporaneous ZIP code characteristics when locating TACs. Consequently, endogenous selection is unlikely to drive any relation between TAC operation and concurrent entrepreneurship.

3.3. Taxpayer assistance and entrepreneurship

Having found support for the conditional exogeneity of TACs, we turn to the main research question: How does taxpayer assistance affect entrepreneurship? Table 3 reports the results of estimating Eq. (1). In odd-numbered columns we include MSA-year fixed effects

[32] An alternative mechanism may be that TACs do not affect marginal potential entrepreneurs directly, but instead affect them only indirectly by providing existing businesses with more time to focus on their core business models, spurring demand for new startups. This alternative mechanism is another way in which TACs can increase new business creation by providing taxpayer assistance.

Table 1

Descriptive statistics and sample selection

This table presents descriptive statistics for the sample and the sample selection criteria. The main sample begins in 2010 and ends in 2015. Panel A reports statistics for variables prior to transformation and Panel B reports statistics for variables after transformation. Panel C presents the sample selection criteria and the number of TACs dropped in each step.

Panel A: Untransformed variables

Variable	N	Mean	Std.	P25	P50	P75
Startups	51,636	168	172	55	114	220
Schedule C income	51,636	28,185	24,890	11,193	20,138	36,411
TAC	51,636	0.032	.	.	.	.
TAC always	51,636	0.026	.	.	.	.
TAC always nearby	51,636	0.331	.	.	.	.
TAC close	51,636	0.001	.	.	.	.
TAC conforming state	51,636	0.024	.	.	.	.
TAC nearby	51,636	0.425	0.721	0	0	1
TAC nonconforming state	51,636	0.008	.	.	.	.
TAC open	51,636	0.0005	.	.	.	.
Number of EITCs	51,636	2448	1995	1060	1861	3180
Number of exemptions	51,636	25,311	13,587	14,680	22,331	32,362
Number of returns	51,636	12,807	6615	7540	11,480	16,450
Number of unemployed	51,636	899	677	414	700	1174
Population	51,636	27,760	14,305	16,415	24,697	35,485
Population 65+	51,636	3528	1924	2081	3128	4547
Population Asian	51,636	1568	2667	202	596	1618
Population Black	51,636	3778	5866	449	1435	4306
Population Native	51,636	161	261	27	76	181
Population Other	51,636	1569	2948	152	469	1488
Population Pacific	51,636	40	106	0	3	26
Population White	51,636	19,519	10,380	11,811	17,514	25,409
Taxable income	51,636	566,973	485,737	239,996	424,816	725,755
Census establishments	5809	6823	10,861	1554	2810	6453

Panel B: Transformed variables

Variable	N	Mean	Std.	P25	P50	P75
ihs(*Startups*)	51,636	5.231	1.377	4.701	5.429	6.087
ihs(*Schedule C income*)	51,636	10.602	0.838	10.016	10.603	11.196
ihs(*Number of EITCs*)	51,636	8.196	0.796	7.659	8.222	8.758
ihs(*Number of exemptions*)	51,636	10.698	0.519	10.287	10.707	11.078
ihs(*Number of returns*)	51,636	10.023	0.510	9.621	10.042	10.401
ihs(*Number of unemployed*)	51,636	7.231	0.744	6.719	7.244	7.761
ihs(*Population*)	51,636	10.801	0.497	10.399	10.808	11.170
ihs(*Population 65+*)	51,636	8.716	0.550	8.334	8.741	9.115
ihs(*Population Asian*)	51,636	6.969	1.686	6.001	7.083	8.082
ihs(*Population Black*)	51,636	7.872	1.622	6.800	7.962	9.061
ihs(*Population Native*)	51,636	4.713	1.833	3.989	5.024	5.892
ihs(*Population Other*)	51,636	6.771	1.838	5.717	6.844	7.998
ihs(*Population Pacific*)	51,636	2.099	2.251	0.000	1.818	3.952
ihs(*Population White*)	51,636	10.412	0.614	10.070	10.464	10.836
ihs(*Taxable income*)	51,636	13.634	0.795	13.082	13.653	14.188
ihs(*Census establishments*)	5809	8.831	1.086	8.042	8.634	9.465

Panel C: Sample selection criteria

	Number of TACs
Total, from the IRS, operating from 2010 to 2015	462
Missing startup data or control variables	(29)
ZIP code population is below 10,000	(75)
Only in operation for 1 year	(20)
Singletons in fixed effects structure	(18)
Final sample	320

Table 2
Determinants of TAC operation
This table presents OLS regressions of TAC operation as a function of ZIP code characteristics. All variables are defined in Appendix A. The *t*-statistics appear in parentheses and are based on standard errors clustered by ZIP code. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two–tail). Sample descriptive statistics are reported in Table 1.

Variable:	*TAC*			
	(1)	(2)	(3)	(4)
ihs(*Number of EITCs*)	0.039***	0.016	0.015	0.017
	(5.06)	(1.34)	(1.20)	(1.34)
ihs(*Number of exemptions*)	−0.192***	−0.007	−0.034	−0.032
	(-7.46)	(-0.17)	(-0.85)	(-0.80)
ihs(*Number of returns*)	0.083***	−0.007	0.005	−0.001
	(3.13)	(-0.27)	(0.22)	(-0.04)
ihs(*Number of unemployed*)0	−0.008*	0.001	0.007*	0.009*
	(-1.76)	(0.38)	(1.67)	(1.65)
ihs(*Population*)	0.059***	0.002	−0.007	−0.005
	(2.73)	(0.15)	(-0.49)	(-0.38)
ihs(*Population 65*+)	−0.001	0.003	0.002	0.002
	(-0.17)	(0.55)	(0.43)	(0.44)
ihs(*Population Asian*)	0.002	−0.001**	−0.001**	−0.001*
	(1.17)	(-2.06)	(-1.98)	(-1.87)
ihs(*Population Black*)	−0.001	0.000	0.001	0.000
	(-0.85)	(0.41)	(0.76)	(0.31)
ihs(*Population Native*)	0.004***	−0.000	−0.000	−0.000
	(4.08)	(-0.44)	(-0.59)	(-1.00)
ihs(*Population Other*)	−0.002	−0.000	−0.000	−0.000
	(-1.55)	(-0.42)	(-0.38)	(-0.89)
ihs(*Population Pacific*)	0.001*	−0.000	−0.000*	−0.000
	(1.65)	(-1.46)	(-1.69)	(-1.60)
ihs(*Population White*)	0.014***	0.005	0.003	0.003
	(3.18)	(1.14)	(0.80)	(0.77)
ihs(*Taxable income*)	0.023***	0.001	0.001	−0.000
	(2.94)	(0.14)	(0.17)	(-0.06)
Year FE	Yes	Yes	No	No
ZIP code FE	No	Yes	Yes	Yes
MSA-year FE	No	No	Yes	No
County-year FE	No	No	No	Yes
Observations	51,636	51,636	51,636	51,636
Adj. R-squared	0.0211	0.922	0.925	0.922

and in even-numbered columns we include the more granular county-year fixed effects. We report results without including controls in columns (1) and (2) to assess how the inclusion of controls affects inferences and to address concerns about potential “bad controls” (Angrist and Pischke, 2009).

Across all four columns of Table 3, we find that TACs associate with increased entrepreneurship (*t*-statistics of 3.02–3.73).[33] In terms of economic magnitudes, the results suggest that the ability to use a TAC associates with a 9 %–16 % increase in entrepreneurial entry, consistent with taxpayer assistance significantly encouraging entrepreneurship.[34] The 90 % confidence interval on *TAC* in the most stringent specification reported in column (4) suggests that TACs associate with a 4 %–13 % increase in entrepreneurial entry. Starting from the median of *Startups*, this corresponds to the creation of 5–15 new businesses (recall from Section 2.1 that the average TAC receives over 13,000 visits each year). Inconsistent with correlated and omitted variables driving these results, including controls for demographic characteristics that the IRS explicitly considers when making TAC operation decisions does not meaningfully affect the coefficient on *TAC*.

In Table 3, the coefficient on *TAC* does attenuate when we include more granular county-year fixed effects in column (4). We offer

[33] Chen and Roth (2024) note that for log and *ihs* transformations of dependent variables that are defined at zero, average treatment effects should not be interpreted as percentage effects. We therefore complement the main results in Table 3 by explicitly calibrating the weight placed on the intensive and extensive margins, as suggested by Chen and Roth (2024). We re-estimate the regressions in Table 3 using the piecewise function from Chen and Roth with varying values of x, where x is the importance of the intensive margin relative to the extensive margin. We find that the piecewise function estimates are most similar to those in Table 3 when $x = 0.5$, suggesting that the *ihs* specification puts a rather small weight on the extensive margin (a change from 0 to 1 startups is valued at the equivalent of a 50 log point change along the intensive margin).

[34] Modelling spillovers requires selecting a boundary within which spillovers are expected and outside which spillovers are not expected. We select 10 miles but note that the results in Table 5 are robust to measuring *TAC nearby* using ZIP codes within 15 or 20 miles, ZIP codes that share borders, and ZIP codes in the same county.

Table 3
Taxpayer assistance and entrepreneurship
This table presents OLS regressions of entrepreneurship as a function of TAC operation. All variables are defined in Appendix A. The *t*-statistics appear in parentheses and are based on standard errors clustered by ZIP code. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two–tail). Sample descriptive statistics are reported in Table 1.

Variable:	*ihs*(*Startups*)			
	(1)	(2)	(3)	(4)
TAC	0.160***	0.089***	0.159***	0.086***
	(3.73)	(3.14)	(3.69)	(3.02)
ihs(*Number of EITCs*)			0.154***	0.183***
			(3.39)	(5.18)
ihs(*Number of exemptions*)			−0.189	−0.220**
			(-1.61)	(-2.34)
ihs(*Number of returns*)			0.346***	0.293***
			(3.35)	(3.33)
ihs(*Number of unemployed*)			−0.079***	0.037*
			(-2.83)	(1.87)
ihs(*Population*)			−0.032	−0.078
			(-0.47)	(-1.57)
ihs(*Population 65+*)			−0.098***	−0.067***
			(-3.65)	(-3.60)
ihs(*Population Asian*)			−0.007*	−0.003
			(-1.77)	(-1.25)
ihs(*Population Black*)			−0.005	0.002
			(-0.81)	(0.54)
ihs(*Population Native*)			0.000	−0.001
			(0.28)	(-0.74)
ihs(*Population Other*)			0.009***	0.005**
			(3.38)	(2.33)
ihs(*Population Pacific*)			−0.001	−0.001*
			(-0.91)	(-1.67)
ihs(*Population White*)			0.078**	0.094***
			(2.45)	(3.81)
ihs(*Taxable income*)			−0.100***	−0.051**
			(-3.58)	(-2.36)
ZIP code FE	Yes	Yes	Yes	Yes
MSA-year FE	Yes	No	Yes	No
County-year FE	No	Yes	No	Yes
Observations	51,636	51,636	51,636	51,636
Adj. R-squared	0.958	0.979	0.959	0.979

two possible explanations for this attenuated coefficient. First, the county-year fixed effects may overcontrol for TAC operations by including a greater proportion of control observations near the TAC that receive positive spillovers from the TAC. Second, the county-year fixed effects may control for an omitted variable that positively varies with both *TAC* and *Startups* at the MSA-year level, but not at the county-year level. While we cannot easily envision such a variable, in Sections 3.4 and 3.5 we explore these two explanations for why including more granular county-year fixed effects attenuates the coefficient on *TAC*.

3.4. *Taxpayer assistance and entrepreneurship, parallel trends*

As discussed in Section 2.1, the IRS considers historical TAC characteristics and location demographics when making TAC operation decisions. Because the IRS uses historical information to select TAC locations, any correlated and omitted variable driving the relation between TACs and contemporaneous entrepreneurship would likely bias the relation several years prior to changes in TAC operation. To explore whether such a correlated and omitted variable drives the results, we examine whether entrepreneurship varies prior to changes in TAC operation decisions.

We re-estimate Eq. (1) after replacing the dependent variable with the inverse hyperbolic sine of the number of startups in each of the prior three years (i.e., $ihs(Startups_{t\text{-}1})$, $ihs(Startups_{t\text{-}2})$, and $ihs(Startups_{t\text{-}3})$). We report results in Table 4. We summarize the results in Fig. 3, which plots the *TAC* coefficient estimate and its 90 % confidence interval along the y-axis against the year in which startup activity was measured along the x-axis. In both Table 4 and Fig. 3, Panel A reports results with MSA-year fixed effects and Panel B reports results with county-year fixed effects. In all panels we find no evidence that startup activity varies as a function of TAC activity prior to changes in TAC operation (i.e., we find consistent evidence of parallel trends in entrepreneurship prior to *TAC*; *t*-statistics on the pre-trend coefficients of −1.46 to 0.65).

The evidence in Table 4 and Fig. 3 that entrepreneurship does not vary prior to changes in TAC operation provides evidence that pre-existing differential trends drive Table 3 results. Moreover, the lack of any pre-trends in entrepreneurship addresses concerns that

Table 4
Taxpayer assistance and entrepreneurship, parallel trends
This table presents OLS regressions of entrepreneurship as a function of TAC operation over time. All variables are defined in Appendix A. The *t*-statistics appear in parentheses and are based on standard errors clustered by ZIP code. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two–tail). Sample descriptive statistics are reported in Table 1. We do not report coefficients on control variables for parsimony. Panel A reports results including MSA-year fixed effects and Panel B reports results including county-year fixed effects.

Panel A: MSA-year fixed effects specification

Variable:	$ihs(Startups_{t-3})$	$ihs(Startups_{t-2})$	$ihs(Startups_{t-1})$	$ihs(Startups)$
	(1)	(2)	(3)	(4)
TAC	−0.039	−0.002	−0.004	0.159***
	(-1.46)	(-0.09)	(-0.18)	(3.69)
Controls	Yes	Yes	Yes	Yes
ZIP code FE	Yes	Yes	Yes	Yes
MSA-year FE	Yes	Yes	Yes	Yes
Observations	25,818	34,424	43,030	51,636
Adj. R-squared	0.977	0.978	0.978	0.959

Panel B: county-year fixed effects specification

Variable:	$ihs(Startups_{t-3})$	$ihs(Startups_{t-2})$	$ihs(Startups_{t-1})$	$ihs(Startups)$
	(1)	(2)	(3)	(4)
TAC	−0.033	0.011	−0.008	0.086***
	(-1.22)	(0.65)	(-0.34)	(3.02)
Controls	Yes	Yes	Yes	Yes
ZIP code FE	Yes	Yes	Yes	Yes
County-year FE	Yes	Yes	Yes	Yes
Observations	25,818	34,424	43,030	51,636
Adj. R-squared	0.977	0.978	0.978	0.979

TAC operation decisions are driven by differential trends in entrepreneurship that would have persisted even in the absence of changes in TAC operation.

3.5. Taxpayer assistance and entrepreneurship, spillover analysis

Having documented evidence that correlated and omitted variables are unlikely to bias the coefficient on *TAC*, we next examine whether the larger coefficient on *TAC* when including MSA-year fixed effects is due to spillovers from TACs to nearby ZIP codes. We define spillovers to include spillovers in taxpayer assistance from nearby TACs and spillovers in economic activity. To capture these spillovers, we re-estimate Eq. (1) after including the number of TACs in ZIP codes within 10 miles of ZIP code *i* (*TAC nearby*).[35] To maximize the ability to detect spillovers, we focus on the MSA-year specification.[36] Table 5 Panel A reports the results of estimating the modified Eq. (1).

The results in Table 5 Panel A provide consistent evidence that TACs increase entrepreneurship in nearby ZIP codes. In particular, the coefficient on *TAC nearby* suggests that TACs relate to an about 4 % increase in entrepreneur entry in nearby ZIP codes, or about one-fourth the effect of having a TAC within the same ZIP code. The coefficient on *TAC* in Panel A is also slightly larger than its counterpart in Table 3, suggesting that spillovers slightly attenuate estimates in Table 3.

The evidence in Table 5 Panel A that the effect of TACs shrinks by three-fourths even for ZIP codes within 10 miles of the TAC's home ZIP code highlights that the effect of TACs rapidly dissipates over relatively short distances. This rapid dissipation suggests that TACs are particularly salient for potential entrepreneurs who live or work in the TAC's home ZIP code, or that potential entrepreneurs face particularly high travel costs. We then examine whether TAC spillover effects are larger for TACs that have been present longer, suggesting salience effects. We repeat the analysis but add indicators for ZIP codes with a TAC present throughout the sample period (*TAC always*), and for ZIP codes with a TAC present nearby throughout the sample period (*TAC nearby always*) and tabulate the results in Panel B. Because *TAC always* and *TAC nearby always* are ZIP code-level variables that do not vary over time, we remove ZIP code fixed effects from the specification. Consequently, we note some caution in interpreting the results.

[35] In the county-year specification the coefficient on *TAC nearby* is small (0.007) and statistically insignificant (*t*-statistic of 0.46), consistent with reduced power to detect spillovers because most ZIP codes in a county with a TAC are within 10 miles of that TAC's ZIP code (i.e., *TAC nearby* = 1 for most observations).

[36] E.g., Feldman et al. (2016), Aghion et al. (2017), Cowx (2021), and Zwick (2021).

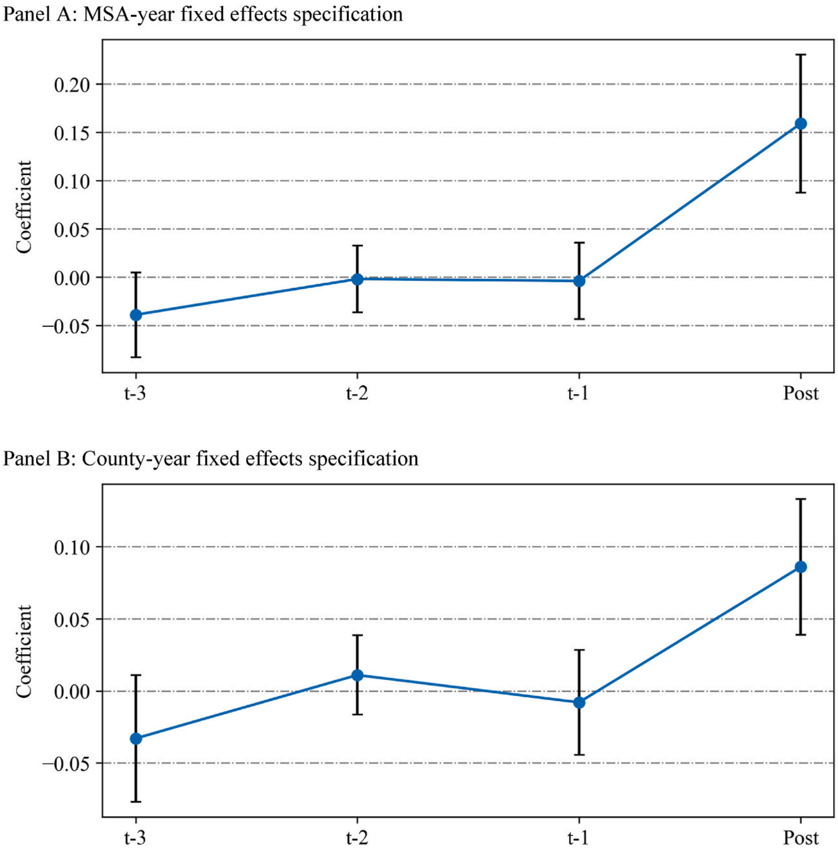

Fig. 3. Tax complexity and entrepreneurship, parallel trends
These graphs present the coefficient and 90 % confidence intervals for *TAC* from the parallel trends test tabulated in Table 4. Panel A reports coefficients from the MSA-year fixed effects specification, and Panel B reports coefficients from the county-year fixed effects specification. The x-axis refers to the year t in which startup activity is measured relative to TAC activity in year $t = 0$.

Consistent with salience affecting the magnitude of spillovers, we find that the coefficient on *TAC nearby always* is 60 % larger than the coefficient on *TAC nearby*, although the difference is not statistically significant. However, we note one wrinkle with this test: only the coefficient on *TAC nearby always* is statistically significant (t-statistic of 2.04). The coefficient on *TAC nearby* is marginally insignificant (t-statistic of 1.58), suggesting that TAC spillovers are more precise and stronger in the time series when including ZIP code fixed effects than in the cross section when excluding them.

3.6. Taxpayer assistance and schedule C income

A potential concern with the prior inferences is that TACs may encourage business registration alone, rather than real business activity. For example, TAC workers may advise individuals asking about taxation on business income they currently report on their personal returns to form corporations, partnerships, or limited liability companies to enjoy the associated (non-tax) legal benefits. In this case, TACs would mechanically increase *Startups* without affecting overall business activity. Although TAC personnel are not supposed to provide non-tax advice, we nonetheless explore the possibility that they do by re-estimating Eq. (1) with *ihs*(*Schedule C income*) as the dependent variable and report results in Table 6. We note that TACs may mechanically decrease Schedule C income by recommending deductions to entrepreneurs, who tend to under-expense on their taxes.[37] However, increases in reported expenses should work against finding a positive effect of TACs on business income.

We find that TACs associate with 7 % increases in *Schedule C income* by (t-statistics of 2.57 and 2.55, with and without MSA-year fixed effects, respectively). The significant relation between TACs and Schedule C income suggests that TACs encourage real business activity, rather than only business registration. That TACs do not merely encourage business registration is perhaps expected given that TAC workers are not tasked with providing general business advice, nor trained in general business law.

[37] We exclude MSA-year fixed effects because on average there are only 1.71 counties within an MSA in the sample. If we estimate the model with MSA-year fixed effects, we find small and insignificant coefficients on *TAC*, likely given the lack of variation. We limit the sample to the counties represented by the ZIP codes in the original analysis.

Table 5
Taxpayer assistance complexity and entrepreneurship, spillover analysis
This table presents OLS regressions of entrepreneurship as a function of TAC operation in the same ZIP code, and TAC operation in ZIP codes within 10 miles. Panel A presents the time-series analysis including ZIP code fixed effects and Panel B presents the cross-sectional analysis excluding ZIP code fixed effects and including *TAC always* and *TAC nearby always*. All variables are defined in Appendix A. The *t*-statistics appear in parentheses and are based on standard errors clustered by ZIP code. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two-tail). Sample descriptive statistics are reported in Table 1. We do not report coefficients on control variables for parsimony.

Panel A: Time-series analysis

Variable:	*ihs*(*Startups*)
	(1)
TAC	0.178***
	(4.07)
TAC nearby	0.037***
	(4.39)
Controls	Yes
ZIP code FE	Yes
MSA-year FE	Yes
Observations	51,636
Adj. R-squared	0.959

Panel B: Cross-sectional analysis

Variable:	*ihs* (*Startups*)
	(1)
TAC always	0.128**
	(2.06)
TAC nearby always	0.040**
	(2.04)
TAC	0.234***
	(4.49)
TAC nearby	0.025
	(1.58)
F-statistic of *TAC nearby always* – *TAC nearby* = 0	0.21
F-statistic *p*-value	0.64
Controls	Yes
ZIP code FE	No
County-year FE	Yes
Observations	51,636
Adj. R-squared	0.923

3.7. Taxpayer assistance and Census establishments

Another potential concern with the prior inferences is that TACs may only cause illegal businesses not reporting their income to the IRS to register and report income, rather than increase real business activity. Although we view this concern as unlikely because the IRS operates TACs independently of its enforcement function, we re-estimate Eq. (1) with *ihs*(*Census establishments*) as the dependent variable and report results in Table 7.

Census establishments is the Quarterly Census of Employment and Wages (QCEW) number of establishments, or physical locations

Table 6
Taxpayer assistance and Schedule C income
This table presents OLS regressions of Schedule C income as a function of TAC operation in the same ZIP code. All variables are defined in Appendix A. The *t*-statistics appear in parentheses and are based on standard errors clustered by ZIP code. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two–tail). Sample descriptive statistics are reported in Table 1. We do not report coefficients on control variables for parsimony.

Variable:	*ihs*(*Schedule C income*)	
	(1)	(2)
TAC	0.073**	0.073**
	(2.57)	(2.55)
Controls	Yes	Yes
ZIP code FE	Yes	Yes
MSA-year FE	Yes	No
County-year FE	No	Yes
Observations	51,510	51,510
Adj. R-squared	0.987	0.987

Table 7
Taxpayer assistance and Census establishments
This table presents OLS regressions of the number of Census establishments as a function of TAC operation in the same county. All variables are defined in Appendix A. The *t*-statistics appear in parentheses and are based on standard errors clustered by county. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two–tail). Sample descriptive statistics are reported in Table 1. We do not report coefficients on control variables for parsimony.

Variable:	*ihs*(*Census establishments*)	
	(1)	(2)
TAC	0.013**	0.014*
	(2.06)	(1.83)
Controls	No	Yes
ZIP code FE	No	No
MSA-year FE	No	No
County-year FE	No	No
County FE	Yes	Yes
Year FE	Yes	Yes
Observations	5809	5809
Adj. R-squared	0.999	0.999

where business is conducted, at the county-year level. Because the QCEW collects business activity from state unemployment and state Census data, QCEW establishment data are less likely to be biased by underreporting to the IRS. Because *Census establishments* is available only at the county level, we additionally modify Eq. (1) by replacing all control variables with their respective county-year values, ZIP code fixed effects with county fixed effects, and MSA/county-year fixed effects with year fixed effects. We further cluster standard errors by county rather than by ZIP code.[38] We find that TACs associate with increases in *Census establishments* by approximately 1.3 % without controls and 1.4 % with controls (*t*-statistics of 2.06 and 1.83).[39] The significant increase again suggests that TACs encourage real business activity.

[38] The average county in our sample spans 8.8 zip codes. In light of Table 5 result that the effect of TACs rapidly dissipates with distance, it is arguably unsurprising that the effect of TACs on establishments at the county-level is smaller than the effect on startups at the zip code-level.
[39] For the 12 ZIP codes in the sample that receive multiple treatments, we keep the first change (i.e., open or close) and drop observations starting the year of the second change, to avoid confounding trends in startup activity from openings followed by closings (or vice versa). For example, if a TAC opened in 2011 and closed in 2013, we mark it as a TAC that opened in 2011 and then drop all related observations from 2013 onward from the regression.

3.8. Taxpayer assistance and entrepreneurship, post trends for TAC openings vs. closings

In this section, we examine how startup activity evolves after a TAC opens or closes. TAC openings should affect entrepreneurship with a delay as potential entrepreneurs become aware of the TAC over time and prepare to launch their business. In contrast, TAC closings should immediately affect entrepreneurship as potential entrepreneurs lose support. To observe differences in the effect between openings and closings, we use a changes specification because the *TAC* indicator in Eq. (1) pools openings and closings. In Table 8, we take the change in all variables of interest, exclude the ZIP code fixed effects since changes control for all time-invariant factors, and estimate the results separately for openings and closings (*TAC open* and *TAC close*).[40] We examine the change in startups in years $t = 0$ through $t = 3$ to document how startup activity evolves after a TAC opens or closes. Panels A and B present results with MSA-year and county-year fixed effects, respectively.

The results suggest that TAC closures decrease entrepreneurship immediately in year $t = 1$, consistent with potential entrepreneurs becoming immediately hindered by the lack of taxpayer assistance in the absence of a TAC. In contrast, TAC openings slowly increase entrepreneurship in year $t = 3$, possibly because it takes time for potential entrepreneurs to learn about a TAC's existence and establish their business. The results are similar in magnitude to the fixed effects results, or are perhaps somewhat larger for closings. However, we note that mapping from changes to fixed effects is not exact since changes specifications require strict timing accuracy while fixed effects specifications are less strict, and since different samples are required to estimate changes at different horizons.

3.9. Taxpayer assistance and entrepreneurship, moderating effect of state tax conformity

We next examine how the conformity of state tax rules to federal taxes moderates the effect of TACs on entrepreneurship. Some states do not have an income tax or enforce tax rules that closely follow the federal rules. In these states, TACs can assist with all, or nearly all, aspects of tax compliance and should therefore affect entrepreneurship more. However, some states' tax rules differ significantly from the federal rules, and in these states TACs are unable to assist with a substantial portion of tax compliance and should therefore affect entrepreneurship less.

To examine the moderating effects of state tax conformity, we estimate Eq. (1) after separating TACs into TACs that operate in states without state income taxes or that begin state tax calculations with federal adjusted gross income (*TAC conforming state*), and TACs that operate in states that build up state income following state-specific rules (*TAC nonconforming state*).[41] We report the results in Table 9. The coefficient on *TAC conforming state* is statistically significant in all columns, while the coefficient on *TAC nonconforming state* is statistically insignificant and about one-fifth the magnitude, suggesting that TACs encourage entrepreneurship mainly in states with no income tax or that closely follow federal income tax rules. Moreover, the difference in coefficients between *TAC conforming state* and *TAC nonconforming state* is significant in all columns, suggesting that state tax conformity significantly moderates the effect of TACs on entrepreneurship.[42]

3.10. Taxpayer assistance and entrepreneurship, stacked differences-in-differences

A potential concern with the staggered differences-in-differences results is that the staggering of treatments can generate biased estimates due to the use of previously treated observations as controls for currently treated observations (e.g., De Chaisemartin and d'Haultfœuille, 2020; Barrios, 2021; Baker et al., 2022). We expect this bias to be minimized in this setting due to the large number of never-treated control observations in the sample and the relatively short panel (De Chaisemartin and d'Haultfœuille, 2020). Nonetheless, we estimate a "stacked" differences-in-differences specification to further rule out the possibility that the staggered differences-in-differences estimates are biased (Gormley and Matsa, 2011).

To estimate a stacked differences-in-differences model, we first remove the 12 ZIP codes in the sample that receive multiple treatments and the 226 ZIP codes that are always treated (i.e., where *TAC* is collinear with the ZIP code fixed effect). We then create datasets in which we pair each ZIP code that experiences a treatment event in year *t* with a cohort of control observations that are not yet treated by year *t* or that are never treated. We then interact the various fixed effects with a cohort indicator that denotes the year of treatment events for each dataset (Baker et al., 2022). Finally, we stack the datasets to form a single sample for which we re-estimate Eq. (1), updated to include the modified fixed effects.

Table 10 reports the results of estimating the updated Eq. (1). The coefficient estimates on *TAC* in columns (1) and (2) are similar to their counterparts in Table 3 (estimates of 0.199 and 0.101, compared to 0.159 and 0.086 in Table 3). Moreover, the coefficient estimates are statistically significant (*t*-statistics of 3.57 and 2.78). We conclude that the traditional staggered differences-in-differences estimates are not meaningfully biased.

4. Conclusion

We examine how taxpayer assistance affects entrepreneurship. Pundits and politicians frequently debate the effect of tax compliance costs on entrepreneurship. We inform this debate by documenting how the ability to use a TAC to navigate the tax system

[40] We identify each type of state via Thomson Reuters: https://www.thomsonreuters.com/en-us/help/ultratax-cs/1040/states/all/agi-and-build-up-states.html, accessed July 1, 2024.

[41] In untabulated tests, we remove states with no income tax from this analysis and find our inferences unchanged.

Table 8
Taxpayer assistance, post trends for TAC openings vs. closings
This table presents OLS regressions of entrepreneurship as a function of TAC openings and closings over time. All variables are defined in Appendix A. The t-statistics appear in parentheses and are based on standard errors clustered by ZIP code. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two–tail). Sample descriptive statistics are reported in Table 1. We do not report coefficients on control variables for parsimony. Panel A reports results including MSA-year fixed effects and Panel B reports results including county-year fixed effects.

Panel A: MSA-year fixed effects specification

Variable:	*ihs*(Δ*Startups*)	*ihs*(Δ$Startups_{t+1}$)	*ihs*(Δ$Startups_{t+2}$)	*ihs*(Δ$Startups_{t+3}$)
	(1)	(2)	(3)	(4)
TAC open	0.086	0.026	−0.072	0.108**
	(1.25)	(0.75)	(-1.38)	(2.40)
TAC close	−0.010	−0.432***	0.030	0.043
	(-0.38)	(-3.35)	(0.57)	(0.91)
ΔControls	Yes	Yes	Yes	Yes
ZIP code FE	No	No	No	No
MSA-year FE	Yes	Yes	Yes	Yes
Observations	43,030	34,424	25,818	17,212
Adj. R-squared	0.735	0.746	0.766	0.788

Panel B: County-year fixed effects specification

Variable:	*ihs*(Δ*Startups*)	*ihs*(Δ$Startups_{t+1}$)	*ihs*(Δ$Startups_{t+2}$)	*ihs*(Δ$Startups_{t+3}$)
	(1)	(2)	(3)	(4)
TAC open	0.053	0.036	−0.080	0.107**
	(0.81)	(1.08)	(-1.59)	(2.25)
TAC close	−0.006	−0.219***	0.032	0.023
	(-0.22)	(-2.99)	(0.59)	(0.52)
ΔControls	Yes	Yes	Yes	Yes
ZIP code FE	No	No	No	No
County-year FE	Yes	Yes	Yes	Yes
Observations	43,030	34,424	25,818	17,212
Adj. R-squared	0.844	0.862	0.888	0.916

Table 9
Taxpayer assistance, moderating effect of state tax conformity
This table presents OLS regressions of startups as a function of TAC operation for states with conforming vs. nonconforming state income tax rules. All variables are defined in Appendix A. The *t*-statistics appear in parentheses and are based on standard errors clustered by ZIP code. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two–tail). Sample descriptive statistics are reported in Table 1. We do not report coefficients on control variables for parsimony.

Variable:	*ihs*(*Startups*)	
	(1)	(2)
TAC conforming state	0.190***	0.102***
	(3.67)	(3.01)
TAC nonconforming state	0.031	0.021
	(0.81)	(0.54)
F-statistic of *TAC conforming state* + *TAC nonconforming state* = 0	11.70	5.79
F-statistic *p*-value	0.001	0.016
Controls	Yes	Yes
ZIP code FE	Yes	Yes
MSA-year FE	Yes	No
County-year FE	No	Yes
Observations	51,636	51,636
Adj. R-squared	0.959	0.979

Table 10
Taxpayer assistance, stacked differences-in-differences
This table presents stacked regressions of entrepreneurship as a function of TAC operation in the same ZIP code. All variables are defined in Appendix A. The *t*-statistics appear in parentheses and are based on standard errors clustered by ZIP code. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels (two–tail). Sample descriptive statistics are reported in Table 1. We do not report coefficients on control variables for parsimony.

Variable:	*ihs*(*Startups*)	
	(1)	(2)
TAC	0.199***	0.101***
	(3.57)	(2.78)
Controls	Yes	Yes
ZIP code-cohort FE	Yes	Yes
MSA-year-cohort FE	Yes	No
County-year-cohort FE	No	Yes
Observations	249,250	249,250
Adj. R-squared	0.958	0.979

affects local entrepreneurship. Consistent with taxpayer assistance affecting entrepreneurship, we find that TACs associate with significant increases in local business creation. We believe that further exploring the consequences of taxpayer assistance is a fruitful avenue for academic research and that policymakers should consider the burdens of compliance when designing tax systems. The results may be of particular interest to policymakers interested in stopping or reversing the decline in traditional business entrepreneurship.

Declaration of competing interest

Stephen Glaeser has no competing interests to disclose.
Daphne Armstrong has no competing interests to disclose.

Appendix A. Variable Definitions

Census establishments	The number of establishments in a county. Obtained from the Quarterly Census of Employment and Wages.
Number of EITCs	The number of tax returns filed that claim the Earned Income Tax Credit. Obtained from IRS Statistics of Income.
Number of exemptions	The number of exemptions on the tax returns filed in the ZIP code. Obtained from IRS Statistics of Income.
Number of returns	The number of tax returns filed in the ZIP code. Obtained from IRS Statistics of Income.
Number unemployed	The number of tax returns filed in the ZIP code that report unemployment compensation. Obtained from IRS Statistics of Income.
Population	The population of the ZIP code. Obtained from Census.
Population 65+	The number of people in the ZIP code who are 65 years of age or older. Obtained from Census.
Population Asian	The number of people in the ZIP code who are Asian. Obtained from Census.
Population Black	The number of people in the ZIP code who are Black. Obtained from Census.
Population Native	The number of people in the ZIP code who are Native American. Obtained from Census.
Population other	The number of people in the ZIP code who are a race other than Asian, Black, Native American, Pacific Islander, or White. Obtained from Census.
Population Pacific	The number of people in the ZIP code who are Pacific Islander. Obtained from Census.
Population White	The number of people in the ZIP code who are White. Obtained from Census.
Schedule C income	The sum of Profit or Loss From Business income reported on tax returns filed in the ZIP code. Obtained from IRS Statistics of Income.
Startups	The number of new business registrations. Obtained from the Startup Cartography Project.
TAC	An indicator variable for whether there is an operational Taxpayer Assistance Center in the ZIP code. Obtained via Freedom of Information Act request to the IRS.
TAC always	An indicator variable for whether there is a Taxpayer Assistance Center in the ZIP code during every year of the sample period.
TAC always nearby	An indicator variable for whether there is a Taxpayer Assistance Center in ZIP codes that are within 10 miles of the focal ZIP code during every year of the sample period.
TAC close	An indicator variable for ZIP codes where a Taxpayer Assistance Center closed in year *t*.
TAC conforming state	An indicator variable for whether there is an operational Taxpayer Assistance Center in the ZIP code, if the ZIP code is in a state that has no state income tax or that begins state tax calculations with federal adjusted gross income.
TAC nearby	The number of Taxpayer Assistance Centers in ZIP codes that are within 10 miles of the focal ZIP code.
TAC nonconforming state	An indicator variable for whether there is an operational Taxpayer Assistance Center in the ZIP code, if the ZIP code is in a state that sets state-specific rules for calculating the personal income tax base.
TAC open	An indicator variable for ZIP codes where a Taxpayer Assistance Center opened in year *t*.
Taxable income	The total of taxable income (in thousands) reported across all tax returns filed in the ZIP code. Obtained from IRS Statistics of Income.

References

Aghion, P., Akcigit, U., Lequien, M., Stantcheva, S., 2017. Tax Simplicity and Heterogeneous Learning. National Bureau of Economic Research. Working Paper w24049.

Allen, E.J., Allen, J.C., Raghavan, S., Solomon, D.H., 2023. On the tax efficiency of startup firms. Rev. Account. Stud. 28 (4), 1887–1928.

Amberger, H., Gallemore, J., Wilde, J., 2023. Corporate tax system complexity and investment. Working paper.

Andrews, R.J., Fazio, C., Guzman, J., Liu, Y., Stern, S., 2022. Reprint of the startup cartography project: measuring and mapping entrepreneurial ecosystems. Res. Pol. 51 (9), 104581.

Angrist, J.D., Pischke, J.S., 2009. Mostly Harmless Econometrics: an Empiricist's Companion. Princeton University Press.

Aobdia, D., Carnes, R.R., Munch, K., 2024. The Role of high-skilled Foreign Accounting Labor in Shaping US Startup Outcomes. Available at SSRN 4686400.

Baker, A.C., Larcker, D.F., Wang, C.C., 2022. How much should we trust staggered difference-in-differences estimates? J. Financ. Econ. 144 (2), 370–395.

Bankman, J., 1994. The structure of silicon valley start-ups. UCLA Law Rev. 41, 1737–1768.

Barrios, J.M., 2021. Staggeringly Problematic: a Primer on Staggered Did for Accounting Researchers. Available at SSRN 3794859.

Barrios, J.M., Choi, J.H., Hochberg, Y.V., Kim, J., Liu, M., 2021. Informing Entrepreneurs: Public Corporate Disclosure and New Business Formation. University of Chicago, Becker Friedman Institute for Economics Working Paper, pp. 2020–2092.

Barrios, J.M., Hochberg, Y.V., Yi, H., 2022. Launching with a parachute: the gig economy and new business formation. J. Financ. Econ. 144 (1), 22–43.

Beers, T., LoPresti, E., San Juan, E., 2012. Factors influencing voluntary compliance by small businesses: Preliminary survey results. 2012 Nation. Taxpay. Advocate Ann. Rep. Congr. 2, 1–70.

Bellemare, M.F., Wichman, C.J., 2020. Elasticities and the inverse hyperbolic sine transformation. Oxf. Bull. Econ. Stat. 82 (1), 50–61.

Benzarti, Y., 2020. How taxing is tax filing? Using revealed preferences to estimate compliance costs. Am. Econ. J. Econ. Pol. 12 (4), 38–57.

Botelho, T.L., Fehder, D., Hochberg, Y., 2021. Innovation-Driven Entrepreneurship. National Bureau of Economic Research. Working Paper w28990.

Bradford, D.F., 1986. Untangling the Income Tax. Harvard University Press.

Burbidge, J.B., Magee, L., Robb, A.L., 1988. Alternative transformations to handle extreme values of the dependent variable. J. Am. Stat. Assoc. 83 (401), 123–127.

Chen, J., Roth, J., 2024. Logs with zeros? Some problems and solutions. Q. J. Econ. 139 (2), 891–936.

Cowx, M.E., 2021. Investment and Tax Incentive Uncertainty: Evidence from the R&D Tax Credit. The Ohio State University, Dissertation.

De Chaisemartin, C., d'Haultføeuille, X., 2020. Two-way fixed effects estimators with heterogeneous treatment effects. Am. Econ. Rev. 110 (9), 2964–2996.

Denes, M., Howell, S.T., Mezzanotti, F., Wang, X., Xu, T., 2023. Investor tax credits and entrepreneurship: evidence from US states. J. Finance 78 (5), 2621–2671.

DePillis, L., 2022. After enduring a pandemic, small businesses face new worries. The New York Times. July 26. Available at https://www.nytimes.com/2022/07/26/business/economy/small-business-recession.html. (Accessed 2 March 2024).

Dunkelberg, W., 2022. Impact of taxes on small business. Forbes. October 6, 2021 (updated April 21, 2022). Available at https://www.forbes.com/sites/williamdunkelberg/2021/10/06/impact-of-taxes-on-small-business/?sh=150746b73250. (Accessed 24 August 2023).

Feldman, N.E., Katuščák, P., Kawano, L., 2016. Taxpayer confusion: evidence from the child tax credit. Am. Econ. Rev. 106 (3), 807–835.

Gentry, W.M., Hubbard, R.G., 2000. Tax policy and entrepreneurial entry. Am. Econ. Rev. 90 (2), 283–287.

Glaeser, S., Omartian, J.D., 2022. Public firm presence, financial reporting, and the decline of US manufacturing. J. Account. Res. 60 (3), 1085–1130.

Goldberg, D.S., 2001. Choice of entity for a venture capital start-up: the myth of incorporation. Tax Lawyer 55 (4), 923–950.

Gormley, T.A., Matsa, D.A., 2011. Growing out of trouble? Corporate responses to liability risk. Rev. Financ. Stud. 24 (8), 2781–2821.

Guzman, J., Stern, S., 2015. Where is silicon valley? Science 347 (6222), 606–609.

Hall, R.E., Woodward, S.E., 2010. The burden of the nondiversifiable risk of entrepreneurship. Am. Econ. Rev. 100 (3), 1163–1194.

Holtz-Eakin, D., Penrod, J.R., Rosen, H.S., 1996. Health insurance and the supply of entrepreneurs. J. Publ. Econ. 62 (1–2), 209–235.

Holtzblatt, J., 2025. Cuts in spending and staff dim hopes for transformational change at the IRS. Tax Policy Center. April 2. Available at https://taxpolicycenter.org/taxvox/cuts-spending-and-staff-dim-hopes-transformational-change-irs. (Accessed 20 June 2025).

Jensen, T.L., Leth-Petersen, S., Nanda, R., 2014. Housing Collateral, Credit Constraints and entrepreneurship-evidence from a Mortgage Reform Working Paper w20583. National Bureau of Economic Research.

Jensen, T.L., Leth-Petersen, S., Nanda, R., 2022. Financing constraints, home equity and selection into entrepreneurship. J. Financ. Econ. 145 (2), 318–337.

Johnson, C.H., 2009. Why do venture capital funds burn research and development deductions? Va. Tax Rev. 29, 29–91.

Johnson, N.L., 1949. Systems of frequency curves generated by methods of translation. Biometrika 36 (1/2), 149–176.

Kamensky, J.M., 2011. Managing the complicated vs. the complex. Business Govern. Magazine (Fall/Winter), 66–67.

Kauffman Foundation, 2018. Breaking barriers: the voice of entrepreneurs. Available at https://www.kauffman.org/wp-content/uploads/2020/06/KauffmanFoundationNationalPolicySurveyofEntrepreneursD22618.pdf. (Accessed 1 September 2023).

Kobe, K., Schwinn, R., 2018. Small Business GDP 1998-2014. Washington, DC: Office of Advocacy. US Small Business Administration.

Lester, R., Olbert, M., 2025. Firms' Real and Reporting Responses to Taxation: a Review. Available at SSRN 4779893.

Matthews, D., 2017. The myth of the 70,000-page federal tax code. Vox. March 29. Available at https://www.vox.com/policy-and-politics/2017/3/29/15109214/tax-code-page-count-complexity-simplification-reform-ways-means. (Accessed 13 July 2023).

Meyer, W., 2011. Where have all the small businesses gone? Forbes (updated August 11). Available at, June 1. https://www.forbes.com/sites/warrenmeyer/2011/06/01/where-have-all-the-small-businesses-gone/?sh=7a01a48b2f7e. (Accessed 2 March 2024).

National Taxpayer Advocate, 2012. Annual report to congress. Available at https://www.taxpayeradvocate.irs.gov/wp-content/uploads/2020/08/Volume-1.pdf. (Accessed 13 February 2025).

National Taxpayer Advocate, 2016. Annual report to congress volume 2. Available at https://www.taxpayeradvocate.irs.gov/wp-content/uploads/2020/08/ARC16_Volume2.pdf. (Accessed 13 February 2025).

Taxpayer Advocate Service, 2018. 2017 annual report to congress. Available at https://www.taxpayeradvocate.irs.gov/wp-content/uploads/2020/10/ARC17_Volume1_MSP_010_TAC.pdf. (Accessed 15 June 2024).

US Chamber of Commerce, 2023. The state of small business now. Available at https://www.uschamber.com/small-business/state-of-small-business-now. (Accessed 2 March 2024).

US Department of the Treasury, 2023. Actions are needed to address inaccurate, incomplete, and inconsistent taxpayer assistance center information provided to taxpayers. Available at https://www.tigta.gov/sites/default/files/reports/2023-08/2023ier006fr.pdf. (Accessed 19 June 2024).

Weber, R., 2015. The effect of tax code complexity on entrepreneurship. J. Priv. Enterprise 30 (2), 83–102.

Zwick, E., 2021. The costs of corporate tax complexity. Am. Econ. J. Econ. Pol. 13 (2), 467–500.